

ModelSim - INTEL FPGA STARTER EDITION 2020.1

EditViewCompileSimulateAddWaveToolsLayoutBookmarksWindowHelp

100 ps

LayoutSimulate

Search:

columnLayoutAllColumns

sim - Default

| Instance          | Design unit    | Design unit type |
|-------------------|----------------|------------------|
| testbench         | testbench(b... | Architecture     |
| hamming_0         | hamming(rtl)   | Architecture     |
| testbench_proc... | testbench(b... | Process          |
| standard          | standard       | Package          |
| textio            | textio         | Package          |
| std_logic_1164    | std_logic_1... | Package          |
| numeric_std       | numeric_std    | Package          |

Processes (Active)

| Name | Type (filtered) |
|------|-----------------|
|------|-----------------|

Objects

| Name       | Value | Type   |
|------------|-------|--------|
| d_signal   | 1110  | Signal |
| p_signal   | 010   | Signal |
| p_expected | 010   | Signal |

second to last iteration  
4 data bits mapping to 010 parity

Wave - Default

/testbench/d\_signal

1110

/testbench/p\_signal

010

/testbench/p\_expec...

010

d signal -> data bits

p\_signal -> xor set output

p\_expected -> manual output using data bit to parity map

| Msgs | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|      | 000  | 101  | 111  | 010  | 011  | 110  | 100  | 001  | 110  | 011  | 001  | 100  | 101  | 000  | 010  | 111  |
|      | 000  | 101  | 111  | 010  | 011  | 110  | 100  | 001  | 110  | 011  | 001  | 100  | 101  | 000  | 010  | 111  |

from data bit value 0 to 15; resulting outputs match mapping of parity bits  
timing offset by 10 ns per test vector, no indication of errors

Now

180 ns

Cursor 1

146.009 ns

0 ns

20 ns

40 ns

60 ns

80 ns

100 ns

120 ns

140 ns

160 ns

180 ns

146.009 ns

Transcript

- Compiling architecture behavioral of testbench

End time: 11:42:11 on Oct 06, 2023, Elapsed time: 0:00:00

Errors: 0, Warnings: 0

sim work.testbench

Start time: 11:42:11 on Oct 06, 2023

loading std.standard

loading std.textio(body)

loading ieee.std\_logic\_1164(body)

loading ieee.numeric\_std(body)

loading work.testbench(behavioral)

loading work.hamming(rtl)

\* Note: Number of errors = 0

Time: 160 ns Iteration: 0 Instance: /testbench

0 ps to 190 ns

/testbench/d\_signal

total runtime 160 ns with 0 errors across test vectors

15 test cases for all combinations of 4 data bit inputs

0 ps to 190 ns

/testbench/d\_signal

2:52 AM

10/7/2023